

Binary Numbers — Short Revision Notes

Analyze Level: Break Information into Parts and Identify Relationships

1. Breaking Down the Binary Number System

Component	What It Represents	Example: 1101 ₂
Digits	The symbols used (0 and 1 only)	1, 1, 0, 1
Place Values	Powers of 2: 1, 2, 4, 8, 16...	8, 4, 2, 1
Position	The place of each digit	3rd, 2nd, 1st, 0th (from right)
Value	Digit x Place Value	8 + 4 + 0 + 1

Relationship Analysis:

Binary:	1	1	0	1
Position:	3rd	2nd	1st	0th
Place:	2 ³	2 ²	2 ¹	2 ⁰
Value:	8	4	0	1

Sum = 8 + 4 + 0 + 1 = **13**

Pattern: Each position is DOUBLE the previous (from right to left)

1 ← 2 ← 4 ← 8 ← 16...

2. Analyzing Place Values in Binary

Position (from right)	Power of 2	Place Value	Decimal System (comparison)
1st	2 ⁰	1	1
2nd	2 ¹	2	10
3rd	2 ²	4	100
4th	2 ³	8	1000
5th	2 ⁴	16	10000
6th	2 ⁵	32	100000
7th	2 ⁶	64	1000000
8th	2 ⁷	128	10000000

Binary place values: 1, 2, 4, 8, 16, 32, 64, 128...

Decimal place values: 1, 10, 100, 1000, 10000...

Both systems follow the same pattern:

- Rightmost place = base⁰ = 1
- Each place to the left = base × previous place
- Binary base = 2, Decimal base = 10

3. Conversion Relationship: Decimal → Binary

Decimal	Binary	Analysis
1	1	$1 = 1 \times 1$
2	10	$2 = 1 \times 2 + 0 \times 1$
3	11	$3 = 1 \times 2 + 1 \times 1$
4	100	$4 = 1 \times 4 + 0 \times 2 + 0 \times 1$
5	101	$5 = 1 \times 4 + 0 \times 2 + 1 \times 1$
6	110	$6 = 1 \times 4 + 1 \times 2 + 0 \times 1$
7	111	$7 = 1 \times 4 + 1 \times 2 + 1 \times 1$
8	1000	$8 = 1 \times 8 + 0 \times 4 + 0 \times 2 + 0 \times 1$

Key Observations:

- Powers of 2 in decimal: 1, 2, 4, 8, 16, 32...
- In binary, these are: 1, 10, 100, 1000, 10000, 100000...
- Each power of 2 is written as 1 followed by n zeros in binary

$$\begin{aligned}
 2^0 &= 1 = 1_2 \\
 2^1 &= 2 = 10_2 \\
 2^2 &= 4 = 100_2 \\
 2^3 &= 8 = 1000_2 \\
 2^4 &= 16 = 10000_2 \\
 2^5 &= 32 = 100000_2
 \end{aligned}$$

4. Conversion Relationship: Binary → Decimal

Binary	Decimal	Analysis
1	1	$1 \times 1 = 1$
10	2	$1 \times 2 = 2$
100	4	$1 \times 4 = 4$
1000	8	$1 \times 8 = 8$
1010	10	$1 \times 8 + 0 \times 4 + 1 \times 2 + 0 \times 1 = 10$
1100	12	$1 \times 8 + 1 \times 4 + 0 \times 2 + 0 \times 1 = 12$
1110	14	$1 \times 8 + 1 \times 4 + 1 \times 2 + 0 \times 1 = 14$

- Each 1 represents a power of 2 that is “turned on”
- Each 0 represents a power of 2 that is “turned off”
- The sum of the “on” powers gives the decimal value

$$\begin{array}{ccccccc}
 1 & 1 & 0 & 1 & & & \\
 \text{ON} & \text{ON} & & \text{ON} & & & \\
 \hline
 1101_2 & = & 8 & + & 4 & + & 0 & + & 1 & = & 13
 \end{array}$$

5. Analyzing Binary Addition — Breaking It Down

Rule	Analysis	Example
$0 + 0 = 0$	No ones to add	$0 + 0 = 0$
$0 + 1 = 1$	One one present	$1 + 0 = 1$
$1 + 1 = 10_2$	Two ones = carry 1, write 0	$1 + 1 = 2_{10} = 10_2$

Analyzing Carrying:

$$\begin{array}{r} 101_2 \\ + 101_2 \\ \hline 1010_2 \end{array}$$

Column Analysis:

Right: $1 + 1 = 2 \rightarrow 10_2$ (write 0, carry 1)

Middle: $0 + 0 + 1(\text{carry}) = 1$

Left: $1 + 1 = 2 \rightarrow 10_2$ (write 0, carry 1)

Extra: 1

Result: 1010_2

Carry Pattern: $1 \rightarrow 0 \rightarrow 1 \rightarrow 1$

Understanding the Pattern:

- When sum = 0 $\rightarrow$ write 0, carry 0
- When sum = 1 $\rightarrow$ write 1, carry 0
- When sum = 2 $\rightarrow$ write 0, carry 1
- When sum = 3 $\rightarrow$ write 1, carry 1

6. Analyzing Binary Subtraction — Breaking It Down

Rule	Analysis	Example
$0 - 0 = 0$	Nothing to subtract	$0 - 0 = 0$
$1 - 0 = 1$	One minus nothing	$1 - 0 = 1$
$1 - 1 = 0$	One minus one	$1 - 1 = 0$
$0 - 1 = \text{Borrow}$	Need to borrow 2 from next column	Borrowing needed

Analyzing Borrowing:

$$\begin{array}{r} 100_2 \\ - 1_2 \\ \hline 011_2 \end{array}$$

Column Analysis:

Right: $0 - 1 \rightarrow$ need to borrow from middle ($0 \rightarrow 2$)

Middle has 0, so it borrows from the left column

Left: $1 \rightarrow 0$, Middle: $0 \rightarrow 2$

Middle lends to right: Middle $2 \rightarrow 1$, Right $0 \rightarrow 2$

Right: $2 - 1 = 1$

Middle: $1 - 0 = 1$

Left: $0 - 0 = 0$

Result: $011_2 = 11_2$

Understanding the Pattern:

- When $0 - 1$, borrow 1 from the next column
- Borrowed 1 = 2 in the current column (since base = 2)
- Continue borrowing if the next column is also 0

7. Analyzing the Relationship: Powers of 2

Power	Value	Binary	Analysis
2^0	1	1	1 zero after 1
2^1	2	10	1 zero after 1
2^2	4	100	2 zeros after 1
2^3	8	1000	3 zeros after 1
2^4	16	10000	4 zeros after 1
2^5	32	100000	5 zeros after 1

$$\begin{aligned}
 2^0 &= 1 = 1_2 \\
 2^1 &= 2 = 10_2 \\
 2^2 &= 4 = 100_2 \\
 2^3 &= 8 = 1000_2 \\
 2^4 &= 16 = 10000_2 \\
 2^5 &= 32 = 100000_2
 \end{aligned}$$

Each power of 2 in binary is written as:
1 followed by n zeros (where n is the power)

Example: $2^4 = 16_{10} = 10000_2$ (1 followed by 4 zeros)

8. Comparing Binary and Decimal Arithmetic

Operation	Decimal	Binary	Analysis
Add 2+2	$2+2 = 4$	$10_2 + 10_2 = 100_2$	Both give same value
Add 5+3	$5+3 = 8$	$101_2 + 11_2 = 1000_2$	Same value in both systems
Add 7+1	$7+1 = 8$	$111_2 + 1_2 = 1000_2$	Carry pattern in binary
Subtract 4-1	$4-1 = 3$	$100_2 - 1_2 = 11_2$	Borrowing in binary

$$\begin{aligned}
 5 + 3 &= 8 \\
 101_2 + 11_2 &= 1000_2
 \end{aligned}$$

Binary addition follows the same rules as decimal:

- Add column by column from right to left
- Carry when sum equals base (2 in binary, 10 in decimal)
- Write the result in base 2

$$\begin{aligned}
 7 + 1 &= 8 \\
 111_2 + 1_2 &= 1000_2
 \end{aligned}$$

All ones + 1 = 1000_2 (like $999 + 1 = 1000$ in decimal)

9. Analyzing Binary Place Values in a Number

Example: 101101_2

Position (from right)	Place Value	Digit	Value
6th (leftmost)	$2^5 = 32$	1	32
5th	$2^4 = 16$	0	0
4th	$2^3 = 8$	1	8
3rd	$2^2 = 4$	1	4
2nd	$2^1 = 2$	0	0
1st (rightmost)	$2^0 = 1$	1	1

Sum: $32 + 0 + 8 + 4 + 0 + 1 = 45_{10}$

$$101101_2 = 45_{10}$$

The 1s are at positions: 6th, 4th, 3rd, 1st

These correspond to: $2^5 + 2^3 + 2^2 + 2^0$

$$= 32 + 8 + 4 + 1 = 45$$

The zeros represent "turned off" powers of 2.

10. Analyzing the Relationship: Number of Digits

Range	Number of Binary Digits	Example
0 - 1	1 digit	0, 1
2 - 3	2 digits	10, 11
4 - 7	3 digits	100, 101, 110, 111
8 - 15	4 digits	1000 to 1111
16 - 31	5 digits	10000 to 11111
32 - 63	6 digits	100000 to 111111

$$\text{Number of digits} = \text{floor}(\log_2(\text{number})) + 1$$

For 8 (binary 1000_2):

$$\log_2(8) = 3, \text{ so digits} = 3 + 1 = 4 \checkmark$$

For 13 (binary 1101_2):

$$\log_2(13) \approx 3.7, \text{ so digits} = 3 + 1 = 4 \checkmark$$

Pattern: The maximum number with n digits $= 2^n - 1$

$$\text{Example: 4 digits max} = 2^4 - 1 = 15 \text{ (1111}_2\text{)}$$

11. Analysis Checklist

When analyzing binary numbers:

- What is the base of the system? (2)
- What are the digits used? (0 and 1)
- What are the place values? (1, 2, 4, 8, 16...)
- What powers of 2 are represented?

When analyzing conversion:

- Decimal → Binary: What remainders are produced?
- Binary → Decimal: What powers of 2 are added?

When analyzing addition:

- Are there any carries?
- Where do carries occur?
- What is the final carry?

When analyzing subtraction:

- Is borrowing needed?
- Where does borrowing occur?
- How many columns are affected?

When analyzing patterns:

- What is the relationship between digits and place values?
- How many digits are needed for each range?
- What is the largest number with n digits?

12. Self-Test — Analysis Questions

No.	Question	Answer & Justification
1	What is the 5th place value in binary?	16 (2^4)
2	Why does $2_{10} = 10_2$?	$2 = 1 \times 2 + 0 \times 1$
3	Why does $8_{10} = 1000_2$?	$8 = 1 \times 8 + 0 \times 4 + 0 \times 2 + 0 \times 1$
4	Max number with 4 binary digits?	15 ($1111_2 = 8+4+2+1$)
5	Why does $111_2 + 1_2 = 1000_2$?	$7+1=8$, all ones become zeros with carry
6	Why is borrowing needed in $100_2 - 1_2$?	0-1 in right column requires borrow
7	What is the pattern of place values?	Each doubles: 1, 2, 4, 8, 16...
8	How many digits to represent 32_{10} ?	6 digits (100000_2)
9	Relationship between 2^4 and 10000_2 ?	$2^4 = 16 = 10000_2$
10	Why does $1010_2 = 10_{10}$?	$1 \times 8 + 0 \times 4 + 1 \times 2 + 0 \times 1 = 10$

13. Key Analytical Insights

1. PLACE VALUES ARE POWERS OF 2:

1, 2, 4, 8, 16, 32, 64, 128...

2. EACH POSITION DOUBLES THE PREVIOUS.

3. POWERS OF 2 = 1 FOLLOWED BY n ZEROS:

$2^n = 1 + n$ zeros in binary.

4. MAX NUMBER WITH n DIGITS = $2^n - 1$.

5. BINARY ADDITION = DECIMAL ADDITION WITH BASE 2:

Carry when sum ≥ 2 . $1+1=10_2$, $1+1+1=11_2$.

6. BINARY SUBTRACTION = DECIMAL SUBTRACTION WITH BASE 2:

Borrow 1 = 2 in current column.

7. BINARY IS BASE 2, DECIMAL IS BASE 10.

Same rules, different base!

8. 1s = "ON" (power of 2 included). 0s = "OFF" (power of 2 not included).

9. LARGER NUMBER = MORE DIGITS OR HIGHER LEFT DIGIT.

10. ANALYZING HELPS UNDERSTAND "WHY" BEHIND CONVERSIONS.

Revision Tip: When analyzing binary numbers, always compare them to the decimal system. The rules are the same — just with base 2 instead of base 10! Understanding the "why" behind conversion and arithmetic operations makes it easier to apply them correctly.